

Asymptotic Relationships

• Running time given in terms of n , where n is size of input

ex. $\text{sort}(x)$, x is list length n , running time in terms of n (eg. n , $\log n$, n^2 , etc.)

• Asymptotic ordering of functions:

$$1 \ll \log n \ll n \ll n \log n \ll n^2 \ll n^2 \log n \ll n^3 \ll 2^n \ll 3^n \ll n!$$

↑
asymptotically smaller (on large inputs)

• Most algebraic rules apply

eg. $f(n) \ll g(n) \rightarrow f(n) \cdot h(n) \ll g(n) \cdot h(n)$

$$n \ll n^2 \rightarrow n \log n \ll n^2 \log n$$

ex. $f(n) = 3^n + 102n^3$ $g(n) = 27n + 2n! + 17$ which asymptotically smaller?

only look at dominant term (largest asymptotically), ignore constants

$$3^n \ll n! \rightarrow f(n) \ll g(n)$$

Big-O

• How long do algorithms/functions take to run?

• How can we analyze running time?

• Looser than asymptotic relationships

• A function $f(n)$ is $O(g(n))$ iff $\exists c, k \in \mathbb{R}^+$ s.t. $0 \leq f(n) \leq c \cdot g(n)$ for $n \geq k$

$$- f(n) \ll g(n) \rightarrow f(n) \text{ is } O(g(n))$$

ex.



$$f(n) = n$$

$$g(n) = n^2$$

$$n \leq c \cdot n^2 \text{ then } n \text{ is } O(n^2) \text{ for } n \geq 1$$

$$\uparrow \\ c=1$$

ex. show $3n^2 + 2n$ is $O(n^3)$ working directly from definition of big O

need to choose constants c & k , show inequality holds true

set $c=2$, need k value s.t. $0 \leq 3n^2 + 2n \leq 2n^3$ for $n \geq k$, what must n be for this to be true?

$$3n^2 + 2n \leq 2n^3$$

$$3n + 2 \leq 2n^2$$

$$\rightarrow 0 \leq 2n^2 - 3n - 2$$

$$0 \leq (2n+1)(n-2)$$

$$\rightarrow n \geq 2$$

↑
choose as k

$$2n+1 \geq 0$$

$$n \geq -\frac{1}{2}$$

If $n \geq 2$, then $n-2 \geq 0$ and $2n+1 \geq 0$. Then, $(n-2)(2n+1) \geq 0$, $2n^2 - 3n - 2 \geq 0$, since $n \geq 2$,

$2n^3 - 3n^2 - 2n \geq 0$, $2n^3 \geq 3n^2 + 2n$, so $3n^2 + 2n$ is $O(n^3)$, we can choose $c=2$, $k=2$.

(k doesn't necessarily have to be smallest possible value)

Big θ

$f(n)$ is $\theta(g(n))$ iff $f(n)$ is $O(g(n))$ and $g(n)$ is $O(f(n))$

ex. Show $2n^2$ is $\theta(n^2)$.

$$\textcircled{1} 2n^2 \text{ is } O(n^2)$$

$$\text{let } c=4$$

$$2n^2 \leq 4n^2$$

$$k=1 \quad (\text{bc true } \forall k)$$

$$\textcircled{2} n^2 \text{ is } O(2n^2)$$

$$c=4$$

$$n^2 \leq 4 \cdot 2n^2 = 8n^2$$

$$\text{choose } k=1$$

Since $2n^2$ is $O(n^2)$ and n^2 is $O(2n^2)$, $2n^2$ is $\theta(n^2)$

Big-O Tutorial Questions

Exponential vs Factorial

- a) Prove that 2^n is $O(n!)$. (Apply the definition of big-O; do not just appeal to our $\ll$ hierarchy, though you may adapt the related proof from the textbook if you need help.)
- b) Prove or disprove the following: If $f(n)$ is $O(2^n)$ and $g(n)$ is $O(n!)$ then $f(n)$ is $O(g(n))$

a) $2^n < c \cdot n!$

We want to show for $c, k \in \mathbb{R}^+$, $0 \leq 2^n \leq c \cdot n!$ for $n \geq k$. If we let $k=4$ and $c=1$, $n \leq 2 + \log_2(n!)$, $\log_2(2^n) \leq \log_2(4n!)$, $2^n \leq 4n!$

b) False, forgets lower bounds. Consider $f(n) = 2^n$ which is $O(2^n)$ and $g(n) = n$ which is $O(n!)$. $f(n)$ is not $O(g(n))$ because 2^n is not $O(n)$

Transitivity of big-O

Prove that if $f(n)$ is $O(g(n))$ and $g(n)$ is $O(h(n))$ then $f(n)$ is $O(h(n))$. (Apply the definition of big-O; do not appeal to general arguments about which functions must grow faster than which others.)

Fix f, g, h , and assume that $f(n)$ is $O(g(n))$ and $g(n)$ is $O(h(n))$. Then by definition of big-O, there are (positive real) k_0, c_0 such that $\forall n \geq k_0, 0 \leq f(n) \leq c_0 g(n)$, and also k_1, c_1 such that $\forall n \geq k_1, 0 \leq g(n) \leq c_1 h(n)$. Now we want to show there are k, c such that $\forall n \geq k, 0 \leq f(n) \leq c h(n)$. Let $k = \max(k_0, k_1)$ and $c = c_0 c_1$. Then we need to show $\forall n \geq \max(k_0, k_1), 0 \leq f(n) \leq (c_0 c_1) \cdot h(n)$. To do this, fix $n \geq \max(k_0, k_1)$. Then we have $0 \leq f(n)$ (since $n \geq \max(k_0, k_1) \geq k_0$), and also:

$$\begin{aligned} f(n) &\leq c_0 g(n) && (\text{since } n \geq \max(k_0, k_1) \geq k_0) \\ &\leq c_0 (c_1 h(n)) && (\text{since } n \geq \max(k_0, k_1) \geq k_1) \\ &= (c_0 c_1) \cdot h(n) && (\text{rearrange}) \end{aligned}$$

Thus, $f(n)$ is also $O(h(n))$.

14.2 Big-O Analysis

For the following problems it is enough to propose values for C and k and briefly justify these by showing that they satisfy the defining inequality for big-o notation.

- (a) Show that $f(n) = n^2 + 8n + 2$ is $\Theta(n^2)$.
- (b) Show that $\frac{x^3+2x}{2x+1}$ is $O(x^2)$.
- (c) Show that $\frac{x^4-1}{x^2+1}$ is $O(x^2)$.
- (d) Show that $2^x + 17$ is $O(3^x)$.

b) $\frac{x^3+2x}{2x+1} \leq x^2$ $x^3+2x \leq 2x^3+x^2$ $x^2+2 \leq 2x^2+x$ $0 \leq x^2+x-2$ $0 \leq (x+2)(x-1)$
 $x \geq -2$ $x \geq 1$

If we set $c=1$, we have to prove $\frac{x^3+2x}{2x+1} \leq x^2$ $\forall x \geq k$. $\frac{x^3+2x}{2x+1} \leq x^2$,
 $x^3+2x \leq 2x^3+x^2$, $k \in \mathbb{R}^+$ and $x \geq k$ so $x^2+2 \leq 2x^2+x$, $0 \leq (x+2)(x-1)$, so
 $x \geq -2$ and $x \geq 1$, use $x \geq 1$ so let $k=1$, $c=1$, which proves $\frac{x^3+2x}{2x+1}$ is $O(x^2)$

d) $2^x + 17 \leq 3^x$ $2^{k-1} \leq 3^{k-1} - 17$ $2^k \cdot (\frac{1}{2}) \leq 3^k (\frac{1}{3}) - 17$ $2^k \leq (\frac{2}{3})(3^k) - 34 \leq 3^k - 17$
 $3^k \leq \frac{3}{2}$

Choose $c=1$ and $k=3$, show $2^x \leq 3^x - 17$ by proving by induction on x . For our base case, because $x \geq k=3$, $x=3$. $2^3 \leq 3^3 - 17$, $8 \leq 10$, so base case holds.

For IH, assume $2^x \leq 3^x - 17$ is true for $x=3, 4, 5, \dots, k-1$. For inductive

step, we have $x=k$ and we need to prove $2^k \leq 3^k - 17$. $2^k \leq 3^k - 17$,
 $2 \cdot 2^{k-1} \leq 3^k - 17$, $2^{k-1} \leq 3^{k-1} - 17$ by IH so $2^k \leq 2(3^{k-1} - 17)$,
 $2^k \leq 2 \cdot 3^{k-1} - 34 < 3 \cdot 3^{k-1} - 34 = 3^k - 34 < 3^k - 17$, thus $2^k \leq 3^k - 17$.